Section 5.3 Geometric probability

几何概率

[AIME-2010-II-Q02] | Part A (Q1-Q5)

Answer: 281

1. The points with nearest-side distance at least t form a central square of side $1-2t$.

2. The desired area, hence probability, is $(3/5)^2-(1/3)^2=56/225$. Boundaries have area zero.

3. The answer is 281.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2017-II-Q03] | Part A (Q1-Q5)

Answer: 409

1. Comparing squared distances gives the half-planes $x > 6$ and $y < (2/5)x + 1$. Intersecting these with the triangle produces vertices $(6, 0), (12, 0), (10, 5), (6, 17/5)$.

2. The quadrilateral area is $15 + 34/5 = 109/5$, by splitting along the diagonal from $(6, 0)$ to $(10, 5)$. The original triangle area is60.

3. The probability is $(109/5)/60 = 109/300$, giving $p + q = 409$. Boundary equalities have area zero.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2020-II-Q02] | Part A (Q1-Q5)

Answer: 171

1. The threshold line is $y = x/2 + 1/16$. To its right of $x = 5/8$ the desired points lie above it; to its left they lie below it. The vertical line itself has area zero.

2. The right trapezoid has area $\frac{3}{8}\frac{5/8+7/16}{2} = 51/256$; the left one has area $\frac{5}{8}\frac{1/16+3/8}{2} = 35/256$.

3. The total area, hence probability, is $86/256 = 43/128$. The answer is 171.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2004-I-Q10] | Part B (Q6-Q10)

Answer: 817

1. With the diagonal from $(0,0)$ to $(36,15)$, centers lie in $[1,35]\times[1,14]$, area 442. The diagonal line is $5x-12y=0$, so avoidance requires $|5x-12y|>13$.

2. One favorable triangle has vertices $(5,1)$, $(35,1)$, $(35,27/2)$, area $375/2$. A half-turn produces an equal region on the other side.

3. Thus probability $375/442$ is reduced, giving 817.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q06] | Part B (Q6-Q10)

Answer: 048

1. The three consecutive arcs between triangle vertices have measures $2x, 2x, 360 - 4x$. A chord misses the triangle exactly when both endpoints lie on the same such arc, apart from zero-probability boundary cases.

2. With $u = x/180$, the miss probability is $2u^2 + (1-2u)^2 = 11/25$. This reduces to $75u^2 - 50u + 7 = 0$, whose roots are $1/5, 7/15$.

3. The corresponding angles $36^\circ, 84^\circ$ both lie in $(0, 90^\circ)$ and form nondegenerate triangles. Their difference is48.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2017-I-Q08] | Part B (Q6-Q10)

Answer: 041

1. Thales theorem places Q, R on the same semicircle with diameter OP and radius100. The chord length is $QR = 200 \sin |a - b|$. Since the angle difference is below75 degrees, the event is equivalent to $|a - b| \leq 30$.

2. In the uniform 75×75 parameter square, the excluded region consists of two triangles of leg45. Its total area is 45^2, so the probability is $1 - 45^2/75^2 = 16/25$. Hence $m + n = 41$.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2023-II-Q06] | Part B (Q6-Q10)

Answer: 035

1. The midpoint always stays in $[0,2]^2$. Failure means its two coordinates both exceed 1, placing it in the missing upper-right square.

2. This requires one point in the upper arm and one in the right arm, probability $2/9$. Conditional on that choice, each coordinate-sum exceeds 2 with probability $1/2$; the two coordinate events are independent.

3. Thus failure has probability $(2/9)(1/2)^2 = 1/18$, and success is $17/18$, yielding 35.

Source / 来源

Section 5.3 Geometric probability

几何概率

[AIME-2003-I-Q11] | Part C (Q11-Q15)

Answer: 092

1. Symmetry about 45° permits restricting to $0 < x \leq 45^\circ$. Here the largest length is $\cos^2 x$, so failure means $\cos^2 x \geq \sin^2 x + \sin x \cos x$.

2. Using double angles, this is $\cos 2x \geq \frac{1}{2} \sin 2x$, or $x \leq \frac{1}{2} \arctan 2$.

3. The two symmetric failure intervals have total degree length $\arctan 2$, so $p = \arctan 2 / 90^\circ$. Thus the requested parameters give $m + n = 2 + 90 = 92$.

Source / 来源